

Methodology Report:

Visualisation & Analysis on Namma Yatri Data

Include your visualisations, analysis, results, insights, and outcomes.

Explain your methodology and approach to the tasks. Add your conclusions to the sections.

Table 1: Data Description

Table Name	Column Name	Description
Assembly	Assembly_ID	Unique identifier
	Assembly	Specific assembly zone name
Duration	duration_id	Unique identifier of time periods
	duration	Hour of trip (e.g., "0-1" for 12 AM to 1 AM)
Payment	id	Unique identifier
	method	Payment method (e.g., Cash, UPI, Credit Card)
Trip Details	tripid	Unique identifier of trips
	loc_from	Source Location code
	searches	Trip request count
	searches_got_estimate	Got an estimated price (1 = user gets an estimate, 0 = does not get an estimate)
	searches_for_quotes	Searched for drivers after estimate (1 - searched, 0 - not searched)
	searches_got_quotes	Got quotes (1 = Driver allotted, 0 = not allotted)
	customer_not_cancelled	Whether customer cancelled or not (1 = Not cancelled)
	driver_not_cancelled	Whether driver cancelled or not (1 = Not cancelled)
	otp_entered	(1 = OTP entered, 0 = not entered)
	end_ride	Whether ride was completed (1 = Completed)
Trips	tripid	Links to Trip Details
	faremethod	Payment method ID, links to Payment table
	fare	Fare amount
	loc_from	Location ID of source
	loc_to	Location ID of destination, links to Assembly table
	driverid	Driver ID
	custid	Customer ID
	distance	Distance in KM from source to destination
	duration	Unique identifier of time periods like duration_id

Points to Note:

1. **Without this methodology document, the other parts of your case study will not be evaluated.**
2. This assignment is different from the ones you have solved before. Make sure that you treat this case study as a storytelling exercise and not an analysis/visualisation one. This will help you be better prepared for the presentations.
3. Once you are done with the analysis and visualisations, there will be many insights at your hand. Make sure that you map the right visuals and takeaways with the right audience since some of these insights might be relevant to one group but not to the other group.
4. **DO NOT** change the text or numbering of any task, as it may cause problems with grading. Write your solutions to a task in the space provided below the respective task.

Tasks to be performed

- Present the overall approach of the analysis.
- Mention the problem statement and the analysis approach briefly.
- To solve a task, you have to create relevant visualisations and derive appropriate insights from the visualisations.
- Add all the plots, insights, calculated field commands, results and outcomes for a task with proper numbering and sequence in the report.
- The scores for all tasks (except conclusions) comprise both analysis work in the visualisation tool and its outcome in the report.
- You will be awarded a score for a task only if the Tableau/PowerBI analysis is correct and is included in the report along with the subsequent insights.
- Finally, draw conclusions based on the analysis.

Scoring:

Report Total Marks: 70

Sections: 3 sections (10 marks + 40 marks + 20 marks)

Analysis and Visualisation

1. Data Preparation

[10 Marks]

1.1. Import and Join Tables Correctly [5 Mark]

- Import the Namma Yatri dataset into Tableau/Power BI.
- Ensure that you correctly join all tables to create a unified dataset for analysis.
- Verify the relationships between different tables and confirm that data from various sources is properly aligned for accurate insights.

Solution:

Tables have been successfully imported in PowerBi.

The relationship between tables is as follows.

- Trips [Duration] - Duration [id]
- Trips [Payment] - Payment [id]
- Trips [loc_from] - Assembly [ID]
- Trips [tripid] - Trip_Details [tripid]
- Trip_Details [loc_to] - Assembly [ID]
- Trip_Details [loc_from] - Assembly [ID]

1.2. Find and Resolve Inconsistencies [5 Marks]

- Identify and resolve any inconsistencies or issues in the dataset that might affect the analysis.
- Clean the data to ensure it is structured properly for analysis, removing any irrelevant, duplicate, or erroneous entries.
- While performing the analysis, create calculated fields as needed to ensure the accuracy and relevance of the insights.

Solution: No inconsistencies or issues are found in the dataset. It has no duplicate value.

Duration data is unavailable for unsuccessful rides.

The calculated fields are as below.

- Average fare per Km
- Average conversion rate
- Average ride per hour
- Percentage of customers got estimate
- Percentage of customer got quotes

- *Percentage of customer searched for quotes*
 - *Percentage of ride after getting quotes*
 - *Percentage of ride canceled by Customer*
 - *Percentage of ride canceled by driver*
 - *Total revenue*
 - *Total successful rides*
 - *Correlation of Total revenue and Average fare per Km on customer ID*
 - *Correlation of Total revenue and Sum of end rides on customer ID*
 - *Correlation of Total revenue and Sum of distance traveled on customer ID*
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2. Exploratory Data Analysis

[40 Marks]

2.1. Classify Variables into Categorical and Numerical [2 Marks]

- Classify all the variables in the dataset into numerical and categorical types.

Solution: The classification of variables is as follows:

Trip

<i>tripid</i>	-Categorical Variable
<i>faremethod</i>	-Categorical Variable
<i>loc_from</i>	-Categorical Variable
<i>loc_to</i>	-Categorical Variable
<i>Driverid</i>	-Categorical Variable
<i>Custid</i>	-Categorical Variable
<i>Duration</i>	-Numerical Variable
<i>Distance</i>	-Numerical variable
<i>Fare</i>	-Numerical variable

Trip_Details

<i>Tripid</i>	-Categorical Variable
<i>loc_from</i>	-Categorical Variable
<i>Searches</i>	- Boolean Variable
<i>searches_got_estimate</i>	- Boolean Variable
<i>searches_for_quotes</i>	- Boolean Variable
<i>searches_got_quotes</i>	- Boolean Variable
<i>customer_not_cancelled</i>	- Boolean Variable
<i>driver_not_cancelled</i>	- Boolean Variable
<i>otp_entered</i>	- Boolean Variable
<i>end_ride</i>	- Boolean Variable

Payments

<i>id</i>	- Categorical Variable
<i>method</i>	- Categorical Variable

Assembly

ID - Categorical Variable

Assembly - Categorical Variable

Duration

id - Categorical Variable

duration - Categorical Variable

2.2. Analyse Ride Demand Over Time [3 Marks]

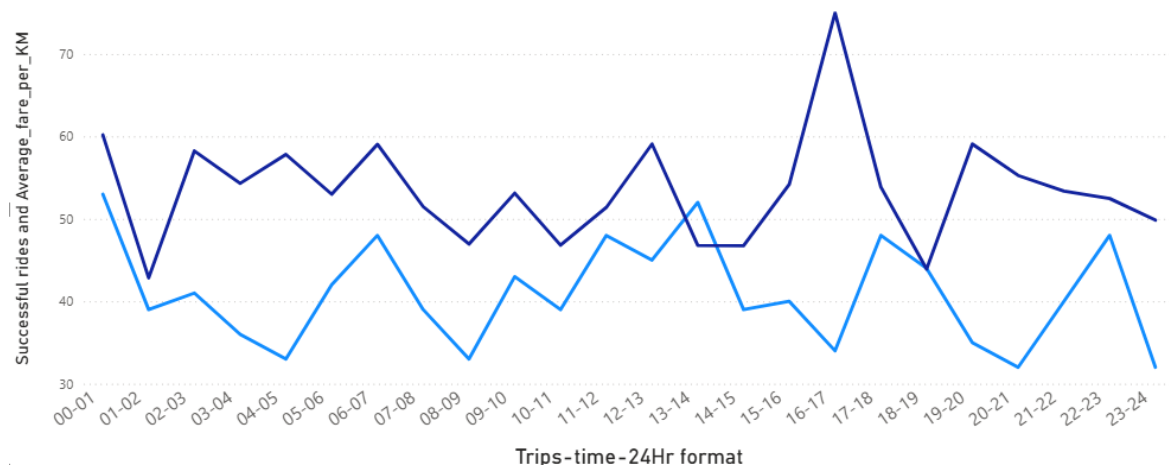
- Explore the distribution of ride demand over time, including trends across different periods.
- Identify the peak demand periods. Choose an appropriate parameter for demand based on your own understanding.

Solution:

1. I will calculate average ride per hour for every time slots.
2. The average ride per hour is 41.
3. Most successful rides are between 22-23,00-01,06-07 and 13-14.
4. The least rides are between 04-05,08-09,16-17,20-21 and 23-24.
5. The average fare per Km is highest for duration 16-17 but has least successful rides.
6. The average fare per Km is lowest for duration 01-02.

Successful rides and Average_fare_per_KM by Trips-time-24Hr format

● Successful rides ● Average_fare_per_KM



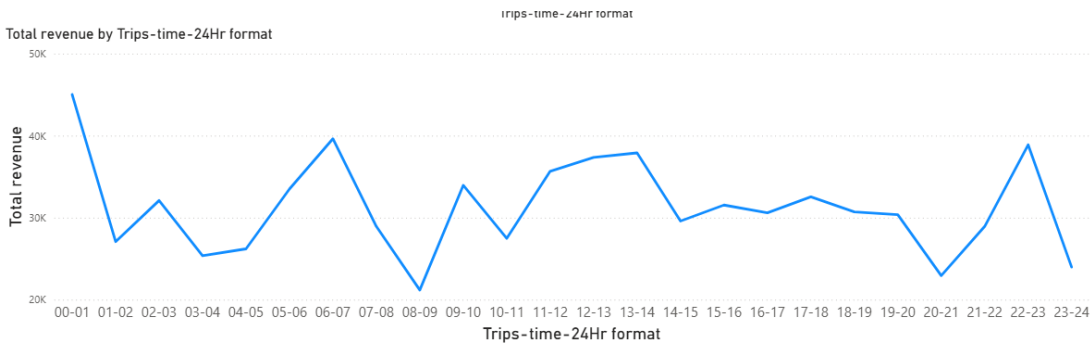
[3 Marks]

- Solution:**

- [3 Marks]**

- Solution:**

- Namma Yatri is losing revenue in peak hours. But it has gained revenue for nightlife travelers and casual afternoon travels.*



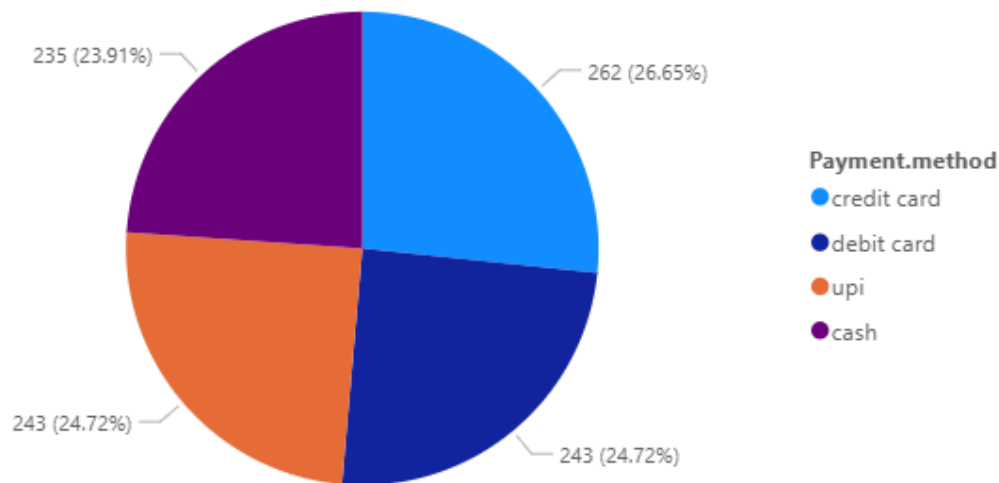
2.5. Examine the Popularity of Different Payment Methods [3 Marks]

- Analyse the distribution of various payment methods used by customers.
- Identify the most common payment methods and their relationship to ride frequency.

Solution:

- *There is not a significant difference in popularity of payment methods.*
- *Credit card is most popular with 26.65% share.*
- *Debit card and UPI have same share 24.72%*
- *Cash is least popular with 23.91%*
- *Customers who use services more frequently prefer to use debit cards and credit cards more frequently.*
- *Customers who use services less frequently prefer to use UPI and Cash more frequently.*

Count of Payment.method by Payment.method



2.6. Identify High-Performing Zones [6 Marks]

Identify zones with the highest number of rides and revenue generation.
Analyse factors contributing to their performance:

- **2.6.1. Rides:** Identify pickup zones with the highest number of trip requests.
[3 marks]

Solution:

The factor affecting revenue regeneration is a good conversion rate above 50% average is 45.49% .

Conversion rate = Successful ride / Total searches.

Following are the topmost requested pickup requests

1. Mahadevpura
2. Ramanagaram
3. Gandhi Nagar
4. Yeshwantpur
5. Dasarahalli

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- **2.6.2. Revenue:** Identify pickup zones generating the highest revenue.
[3 marks]

Solution:

The following are the topmost revenue generating zones.

- | | |
|------------------------|----------|
| 1. Bangalore South | (30,295) |
| 2. Yeshwantpur | (29,035) |
| 3. Hebbal | (28,097) |
| 4. Rajarajeshwarinagar | (26,509) |
| 5. Ramanagaram | (26,225) |

2.7. Analyse Ride Time Periods Across Zones [4 Marks]

Compare the trip trends for different time periods across pickup zones.

Solution:

For most revenue generating time zones following are the most active pickup zones.

1. Basavanagudi,
2. Gandhi Nagar,
3. Bangalore South,
4. Chickpet
5. Shivaji Nagar

For least revenue generating time zones following are the most active pickup zones

1. Ramanagaram
2. Govindraj Nagar
3. Rajarajeshwarinagar
4. Yeshwantpur
5. Bangalore South

During other times the following remain most active.

1. Yeshwantpur
2. Bommanahalli
3. Hebbal
4. Malleshwaram

5. Dasarahalli

2.8. Top Zones with Highest Trip Volume [3 Marks]

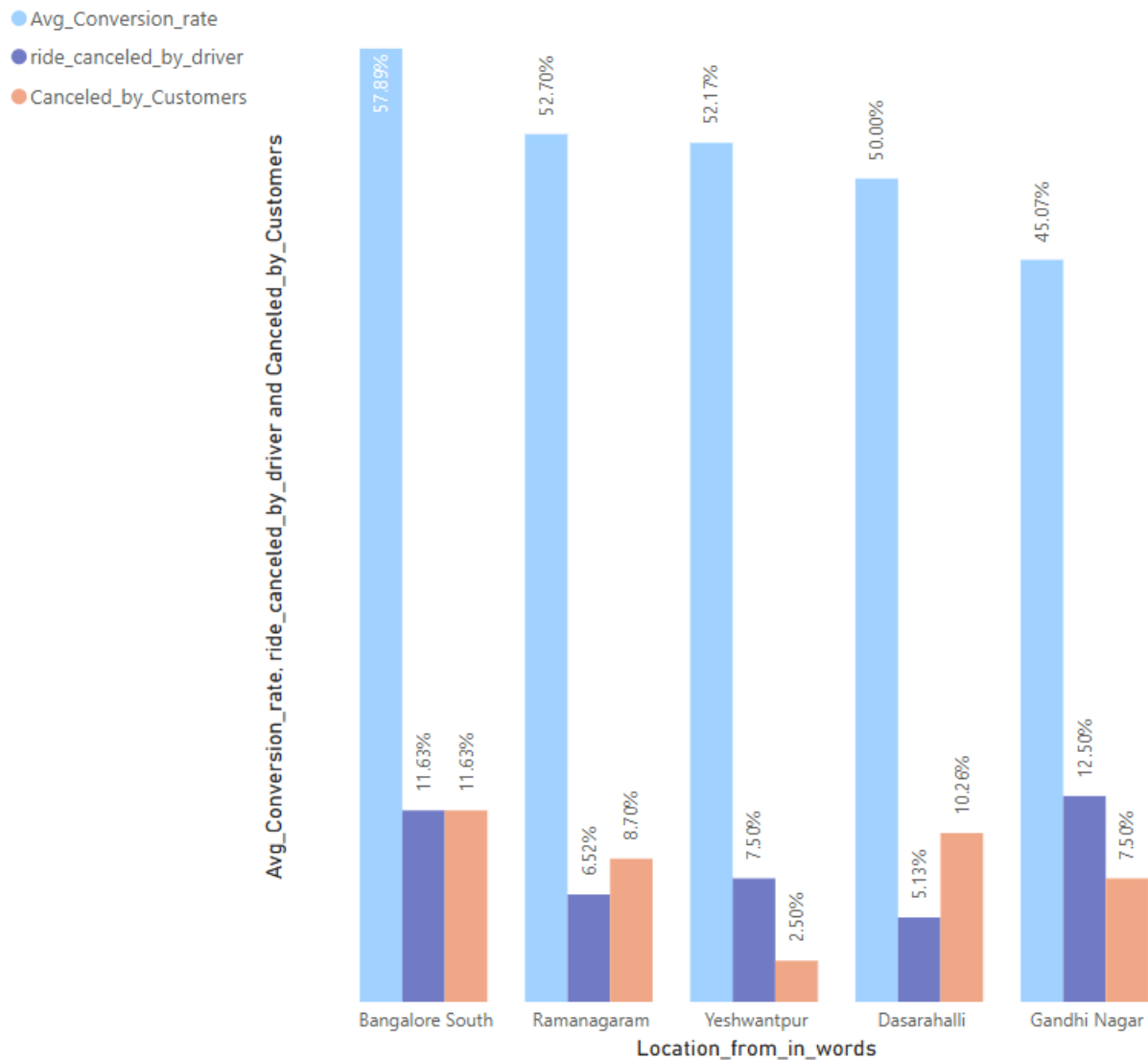
1. Identify the top 5 pickup zones with the highest total number of completed trips.
2. Analyse factors contributing to the higher number of trips.

Solution:

The top 5 pickup zones with highest number of completed trips are:

1. Ramanagaram
2. Yeshwantpur
3. Bangalore South
4. Dasarahalli
5. Gandhi Nagar

The factors affecting higher trips are good conversion rate and low level of cancelation from customer and driver.



2.9. Basic Analytical Tasks [8 Marks]

1. 2.9.1

What are the percentages of cancellations and successful rides by both driver and customer? [3 marks]

Solution:

Drivers have canceled 10.73% of total ride

Customers have canceled 12.29% of rides after getting quotes.

The successful rides after getting quotes are 76.98%

2. **2.9.2**

Analyse the percentage of people who completed trips after searching for quotes. Visualise the variation of this ratio by time periods.
[5 marks]

Solution:

The percentage of customers completed rides after searching for quotes is 67.56%.

The duration is provided only for successful trips, so visualization of the above ratio based on time is not possible.

2.10. Create a Parameter and Use Filters [5 Marks]

1. Create a parameter and use it as a filter on an appropriate subset of the data to interactively analyse and visualise different subsets of the data.
2. Explain your choice of filter and insights drawn from this step.

Solution:

I have created parameters like

- ***Average Conversation rate***
- ***Percentage of customer got estimate***
- ***Percent of customer searched for quotes***
- ***Percent of customer got quotes***
- ***Percentage of rides canceled by customer***
- ***Percentage of rides canceled by drivers***

I will filter the above parameters based on the source location.

This will allow me to analyze and visualize conversation rate, cancelation rate, issues related to estimation of fare and customer getting quotes based on source location of customer. This will help for better allocation of resources.

3. Conclusion

[20 Marks]

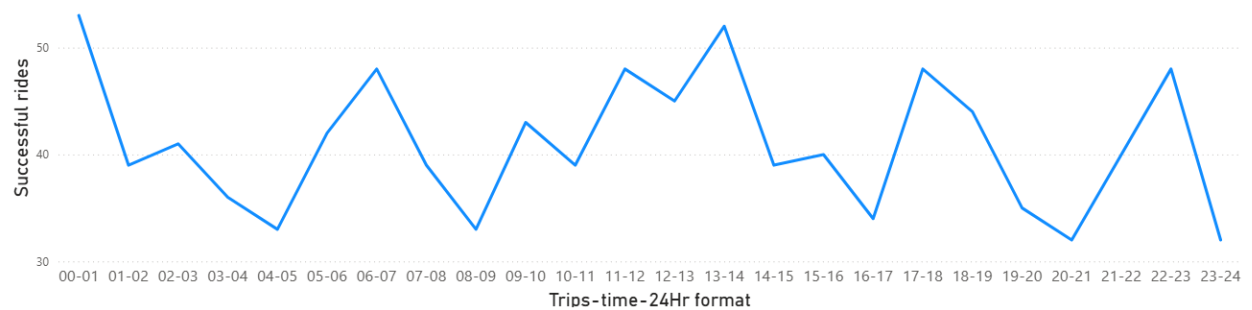
3.1. Recommendations for Operational Efficiency [10 Marks]

- Based on your findings from the analysis, provide recommendations on how Namma Yatri can optimise its operations.
- This could include strategies for improving resource allocation, reducing cancellations, or optimising ride durations.
- Add supporting dashboards.

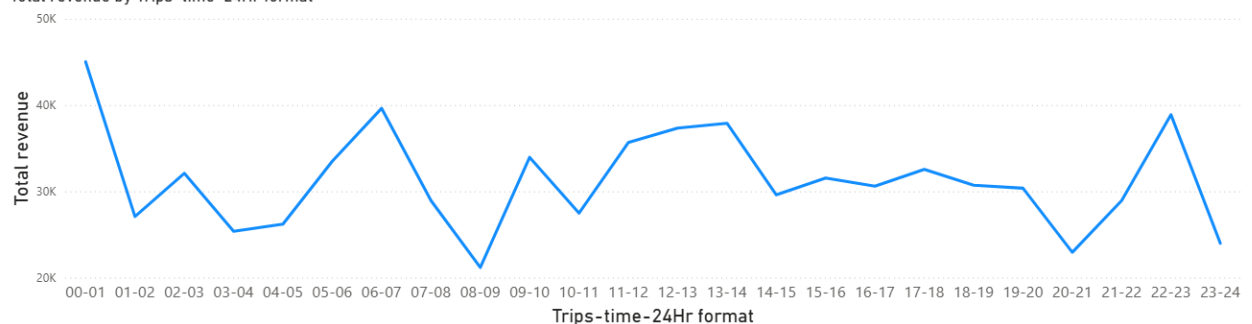
Solution:

1. *There is a drop in rides in the morning (8 am to 11 am) and evening (5 am to 8 am) rush hour. Means office travelers, schools or college students don't use our services often. Also, there is a drop in revenue.*
2. *There are spikes in late night travels 10pm to 11pm and 12am to 1 am probably people who like to enjoy night life.*
3. *There are spikes in travel and revenue from 11 am to 2 pm.*

Successful rides by Trips-time-24Hr format



Total revenue by Trips-time-24Hr format



Time series analysis on total successful ride and total revenue

Taking into consideration the above points, peak time for Namma Yatri trips and revenue wise is 11 am to 2 pm, 10pm to 11pm and 12am to 1am.

1) To increase the operation efficiency, we must know most pick up location to destination peak hours

Time slot	Source of travel	Count of successful rides
11am to 12pm	Chamrajpet	5
11am to 12pm	Yeshwantpur	5
12pm to 01pm	Vijay Nagar	4
12pm to 01pm	Chamrajpet	3
12pm to 01pm	Hebbal	3
12pm to 01pm	Kanakapura	3
12pm to 01pm	Rajarajeshwarinagar	3
12pm to 01pm	Shivajinagar	3
01pm to 02pm	Bangalore South	4
01pm to 02pm	Byatarayanapura	4
01pm to 02pm	Devanahalli	3
01pm to 02pm	Jayanagar	3
01pm to 02pm	Magadi	3
01pm to 02pm	Mahadevapura	3
01pm to 02pm	Shivajinagar	3
01pm to 02pm	Vijay Nagar	3
10pm to 11pm	Other Assemblies	4
10pm to 11pm	Pulakeshinagar	3
10pm to 11pm	C. V. Raman Nagar	3
10pm to 11pm	Magadi	3
12pm to 01am	Dasarahalli	4
12pm to 01am	Chamrajpet	3
12pm to 01am	Gandhi Nagar	3
12pm to 01am	Hoskote	3
12pm to 01am	Padmanabhanagar	3
12pm to 01am	Sarvagnanagar	3

Above are the most pick up location for time zones. To improve operation efficiency,

we must allocate most of the resources to the above region at that time.

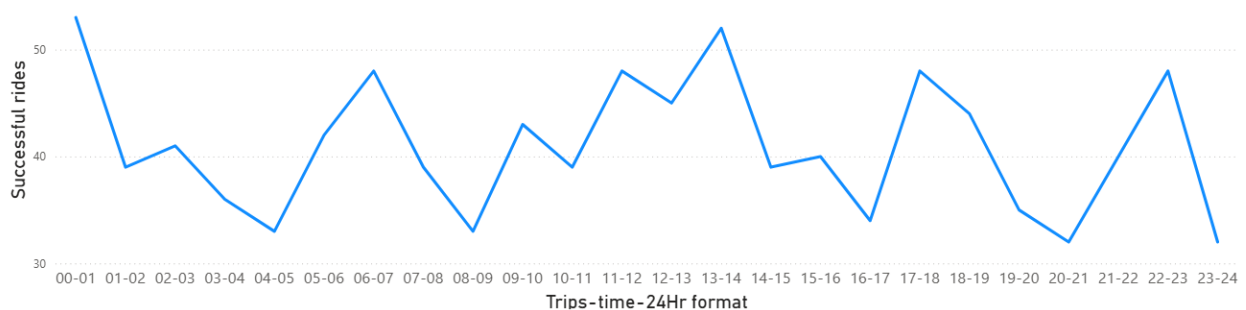
3.2. Marketing and Operational Strategy Improvements [10 Marks]

- Suggest improvements to Namma Yatri's marketing or operational strategies based on your analysis.
- Recommendations could involve promotional efforts, driver incentives, or regional targeting to increase customer satisfaction and service efficiency.
- Add supporting dashboards.

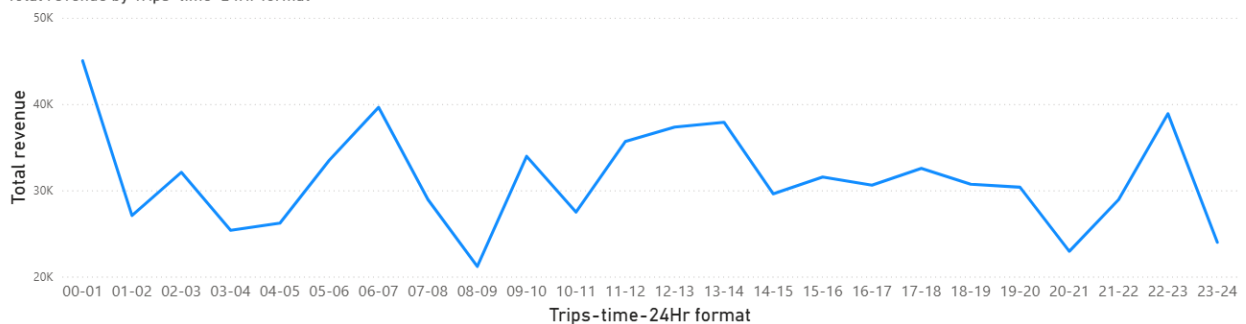
Solution:

1. Namma Yatri is losing revenue in peak hours
2. Namma Yatri successful rides are dropping in peak hours.

Successful rides by Trips-time-24Hr format



Total revenue by Trips-time-24Hr format



3. Namma Yatri must promote itself as reliable, punctual and affordable service provider. To increase the successful rides in peak hours
4. On average, 10.73% rides are canceled by drivers and 12.29% canceled by customer.
5. On location like Mahadevpura, Anekal, Magadi the cancelation by drivers is above 20% we must investigate this issue and take action like increasing drivers incentive or increase its efficiency
6. On location like C. V. Raman Nagar, Doddaballapur and Jayanagar the cancelation is above 20%. The cancelation for C. V. Raman Nagar is 30%.

Customer satisfaction must improve in this location by increasing operation efficiency.

